LABORATÓRIO DE SERVIÇOS DE REDES DE COMPUTADORES
Relatório aula prática - Servidor DHCP

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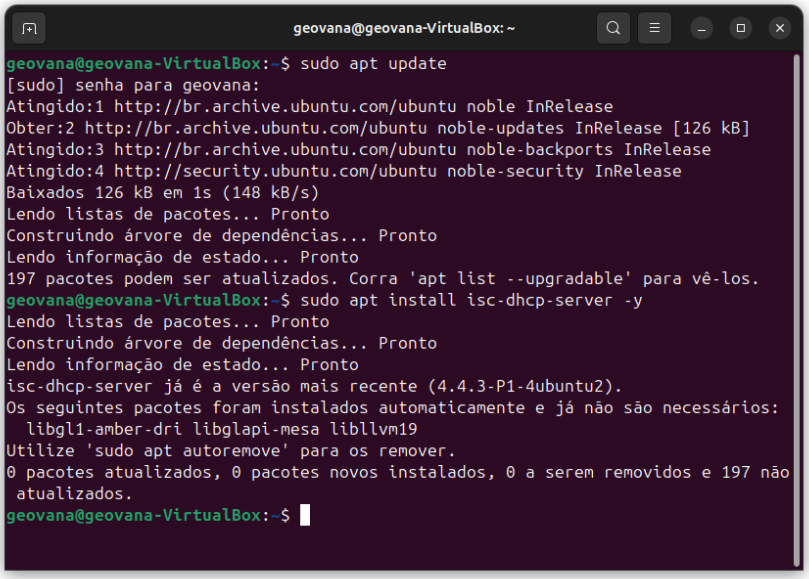
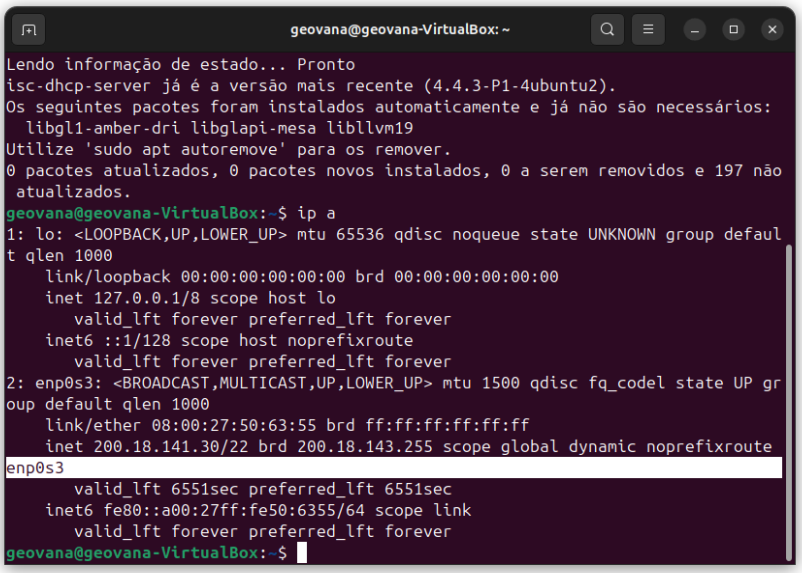
Seguindo as instruções propostas:
1. Configurar o Servidor DHCP (VM1)
1.1. Instalar o servidor DHCP
sudo apt update
sudo apt install isc-dhcp-server -y

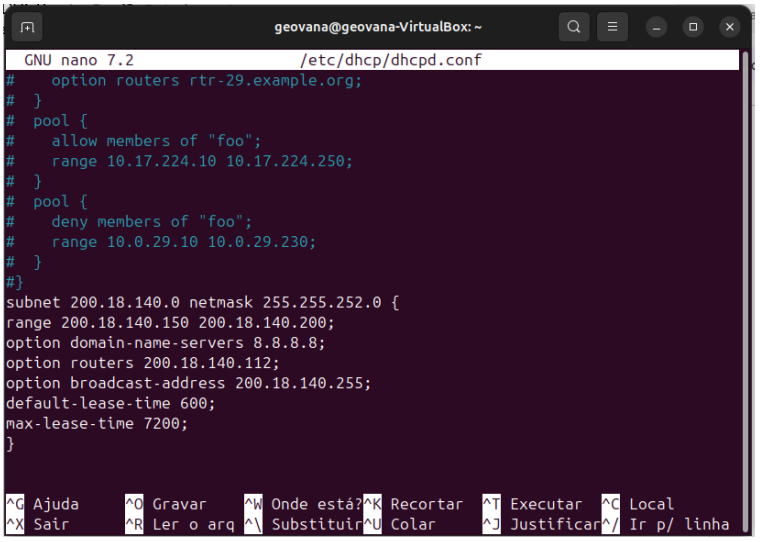


1.2. Identificar a interface de rede: *ip a*
• Nome da interface: enp0s3.

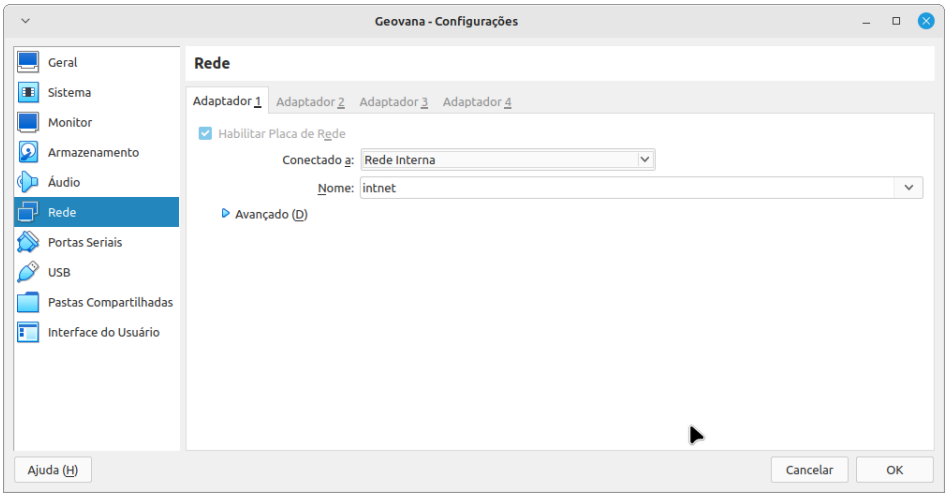


1.3. Editar o arquivo de configuração do DHCP
sudo nano /etc/dhcp/dhcpd.conf

Configuração:
subnet 200.18.140.0 netmask 255.255.252.0 {
range 200.18.140.150 200.18.140.200;
option domain-name-servers 8.8.8.8;
option routers 200.18.140.112;
option broadcast-address 200.18.140.255;
default-lease-time 600;
max-lease-time 7200;
}



1.4. Configurar a VM como rede interna e reinicie a placa de rede



1.5. Configurar a interface de rede com IP estático
sudo nano /etc/netplan/01-netcfg.yaml

```
geovana@geovana-VirtualBox: ~
  GNU nano 7.2                    /etc/netplan/01-netcfg.yaml
network:
  version: 2
  renderer: networkd
  ethernets:
    enp0s3:
      dhcp4: no
      addresses:
        - 200.18.140.1
      nameservers:
        addresses:
          - 8.8.8.8
          - 1.1.1.1

                                [ Escritas 12 linhas ]
^G Ajuda      ^O Gravar     ^W Onde está? ^K Recortar   ^T Executar   ^C Local
^X Sair       ^R Ler o arq  ^\ Substituir ^U Colar      ^J Justificar ^/ Ir p/ linha
```

Aplique as configurações:
sudo netplan apply

1.6. Informar ao serviço qual interface usar
sudo nano /etc/default/isc-dhcp-server

Modifique a linha: INTERFACESv4="enp0s3" (nome da rede identificada)

```
geovana@geovana-VirtualBox: ~
  GNU nano 7.2                /etc/default/isc-dhcp-server *
# Defaults for isc-dhcp-server (sourced by /etc/init.d/isc-dhcp-server)

# Path to dhcpd's config file (default: /etc/dhcp/dhcpd.conf).
#DHCPDv4_CONF=/etc/dhcp/dhcpd.conf
#DHCPDv6_CONF=/etc/dhcp/dhcpd6.conf

# Path to dhcpd's PID file (default: /var/run/dhcpd.pid).
#DHCPDv4_PID=/var/run/dhcpd.pid
#DHCPDv6_PID=/var/run/dhcpd6.pid

# Additional options to start dhcpd with.
#       Don't use options -cf or -pf here; use DHCPD_CONF/ DHCPD_PID instead
#OPTIONS=""

# On what interfaces should the DHCP server (dhcpd) serve DHCP requests?
#       Separate multiple interfaces with spaces, e.g. "eth0 eth1".
INTERFACESv4="enp0s3"
INTERFACESv6=""

^G Ajuda      ^O Gravar     ^W Onde está? ^K Recortar   ^T Executar   ^C Local
^X Sair       ^R Ler o arq  ^\ Substituir ^U Colar      ^J Justificar ^/ Ir p/ linha
```

Inicie o serviço:
sudo systemctl restart isc-dhcp-server

sudo systemctl status isc-dhcp-server

```
geovana@geovana-VirtualBox: ~
geovana@geovana-VirtualBox:~$ sudo systemctl restart isc-dhcp-server
geovana@geovana-VirtualBox:~$ sudo systemctl status isc-dhcp-server
● isc-dhcp-server.service - ISC DHCP IPv4 server
     Loaded: loaded (/usr/lib/systemd/system/isc-dhcp-server.service; enabled; >
     Active: active (running) since Mon 2025-09-15 10:39:27 -03; 6s ago
       Docs: man:dhcpd(8)
   Main PID: 9326 (dhcpd)
      Tasks: 1 (limit: 12437)
     Memory: 3.7M (peak: 4.5M)
        CPU: 25ms
     CGroup: /system.slice/isc-dhcp-server.service
             └─9326 dhcpd -user dhcpd -group dhcpd -f -4 -pf /run/dhcp-server/d>

set 15 10:39:27 geovana-VirtualBox sh[9326]: Wrote 0 leases to leases file.
set 15 10:39:27 geovana-VirtualBox dhcpd[9326]: PID file: /run/dhcp-server/dhcp>
set 15 10:39:27 geovana-VirtualBox dhcpd[9326]: Wrote 0 leases to leases file.
set 15 10:39:27 geovana-VirtualBox dhcpd[9326]: Listening on LPF/enp0s3/08:00:2>
set 15 10:39:27 geovana-VirtualBox sh[9326]: Listening on LPF/enp0s3/08:00:27:5>
set 15 10:39:27 geovana-VirtualBox sh[9326]: Sending on   LPF/enp0s3/08:00:27:5>
set 15 10:39:27 geovana-VirtualBox sh[9326]: Sending on   Socket/fallback/fallb>
set 15 10:39:27 geovana-VirtualBox dhcpd[9326]: Sending on   LPF/enp0s3/08:00:2>
set 15 10:39:27 geovana-VirtualBox dhcpd[9326]: Sending on   Socket/fallback/fa>
set 15 10:39:27 geovana-VirtualBox dhcpd[9326]: Server starting service.
lines 1-21/21 (END)
```

2. Configurar o Cliente DHCP (VM2 – Ubuntu pode ser inicializado como teste)

Configure como rede interna (de mesmo nome da VM1), desative e ative (reinicie) a rede e, se o servidor (VM1) estiver rodando, o IP do cliente (VM2) deve estar no intervalo (range) definido pela configuração do servidor.

```
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr
oup default qlen 1000
    link/ether 08:00:27:2b:f8:60 brd ff:ff:ff:ff:ff:ff
    inet 200.18.140.151/22 brd 200.18.143.255 scope global dynamic noprefixroute
 enp0s3
       valid_lft 555sec preferred_lft 555sec
    inet6 fe80::a00:27ff:fe2b:f860/64 scope link
       valid_lft forever preferred_lft forever
ubuntu@ubuntu:~$ ping
ping: usage error: Destination address required
ubuntu@ubuntu:~$ ping 200.18.140.1
PING 200.18.140.1 (200.18.140.1) 56(84) bytes of data.
64 bytes from 200.18.140.1: icmp_seq=1 ttl=64 time=0.906 ms
64 bytes from 200.18.140.1: icmp_seq=2 ttl=64 time=1.05 ms
64 bytes from 200.18.140.1: icmp_seq=3 ttl=64 time=1.06 ms
64 bytes from 200.18.140.1: icmp_seq=4 ttl=64 time=1.12 ms
```

Máquina Virtual em modo teste - Rede interna e método IPV4 DHCP